

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Solution Architecture

Online ComplaintRegistrationandManagementSystems — MERN
Stack E-commerce Application

Phase: Project Design Phase

Document Version: 1.1

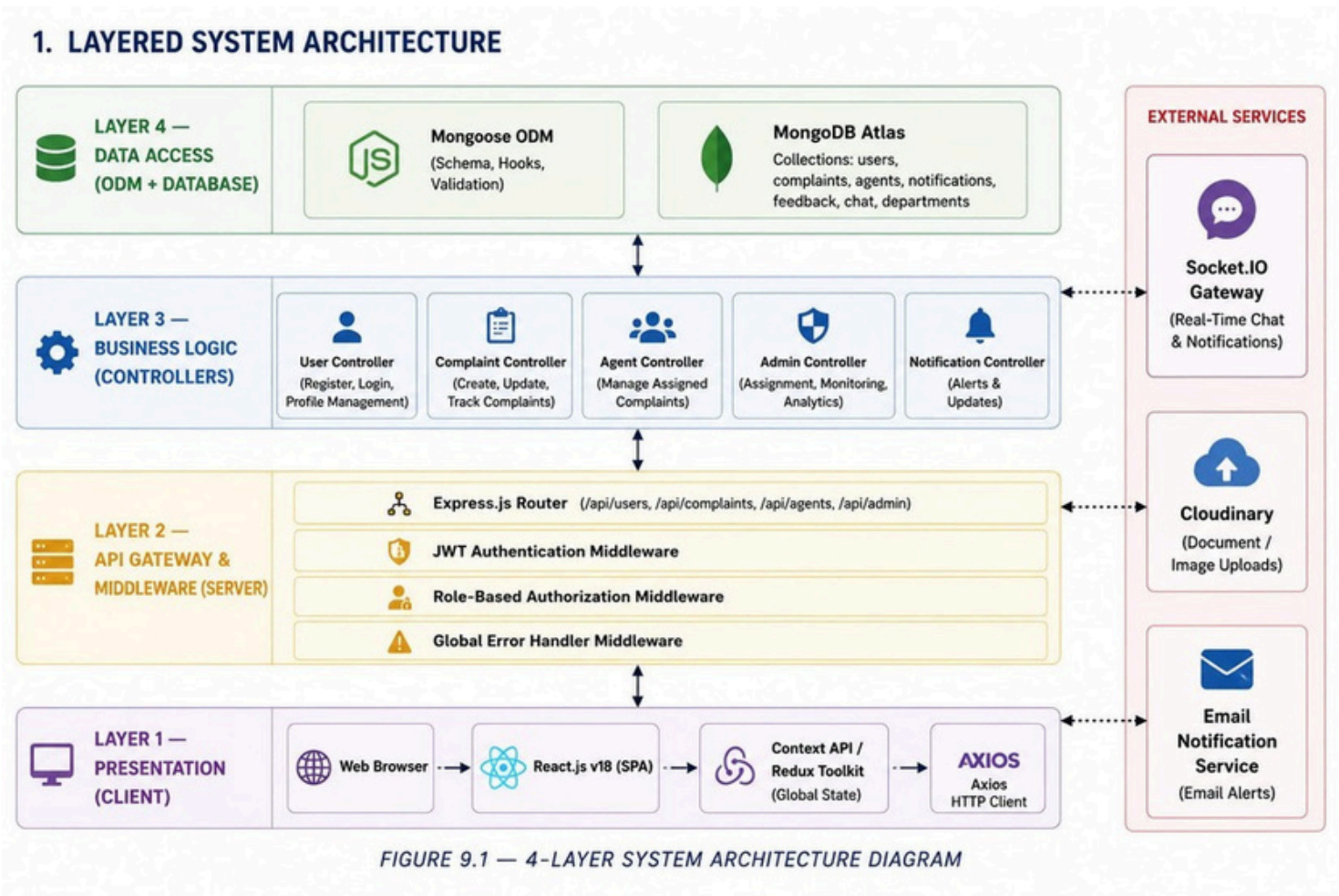
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Classification: Project Documentation — For Academic Submission

Solution Architecture

1. Layered System Architecture



2. MVC Pattern Mapping

MVC Layer	MERN Component	Responsibilities
View	React.js (Frontend SPA)	Renders UI components, manages state, sends API requests, updates dashboards and complaint information
Controller	Express.js Controllers	Receives validated requests, processes complaint logic, handles assignments, returns JSON responses
Model	Mongoose Schemas & Models	Defines complaint, user, agent, notification, and feedback schemas; validates and stores data in MongoDB Atlas

3. UML Class Diagram – Mongoose Models

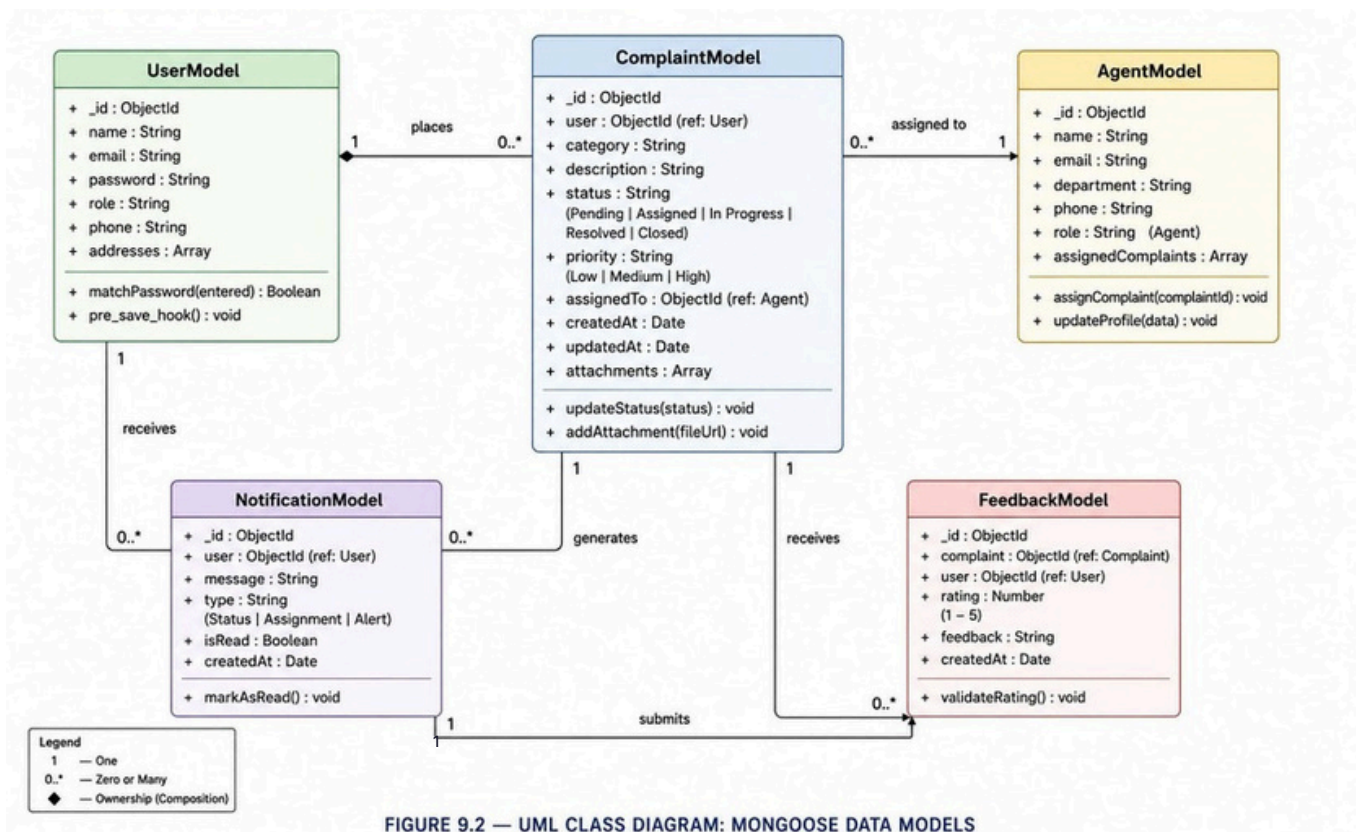


FIGURE 9.2 — UML CLASS DIAGRAM: MONGOOSE DATA MODELS

4. Deployment Topology

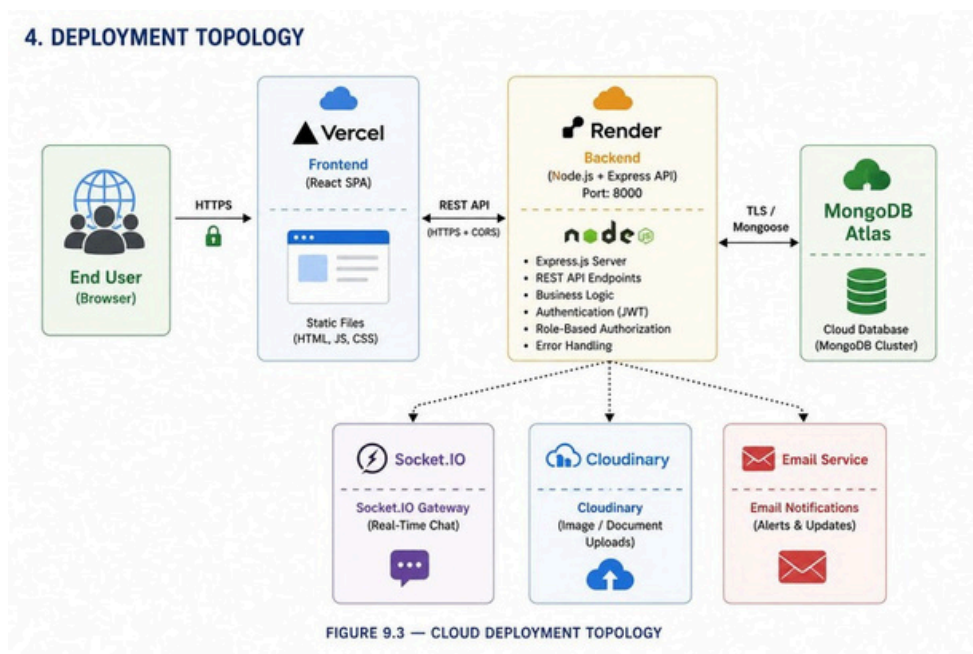


FIGURE 9.3 — CLOUD DEPLOYMENT TOPOLOGY

5. Deployment URL Reference

Component	Platform	URL
Frontend (React SPA)	Vercel	https://vercel.com/espadass/online-complaint-management/settings/environment-variables
Backend (Node.js API)	Render	https://dashboard.render.com/web/srv-d8puhfrtqb8s738jctq0/env
Database	MongoDB Atlas	Cloud Hosted Cluster (Shared Sandbox Instance Tier)
Real-Time Communication	Socket.IO	Integrated Connection Protocol Layer with API Endpoint

6 .Architectural Benefits

The OCRMS architecture provides a secure, scalable, and maintainable foundation for complaint management. Its layered MERN-based design ensures efficient data handling, real-time communication, centralized complaint tracking, and role-based access control. Cloud deployment enhances reliability and accessibility, while the modular structure supports future enhancements such as AI integration, mobile applications, and advanced analytics. Overall, the architecture improves system performance, user experience, and operational efficiency.

7 .Conclusion

The OCRMS architecture provides a secure, scalable, and future-ready foundation for digital complaint management. Its modular design, cloud deployment strategy, and real-time capabilities ensure efficient complaint resolution, improved stakeholder satisfaction, and long-term adaptability to evolving organizational requirements.